Backward value iteration – example

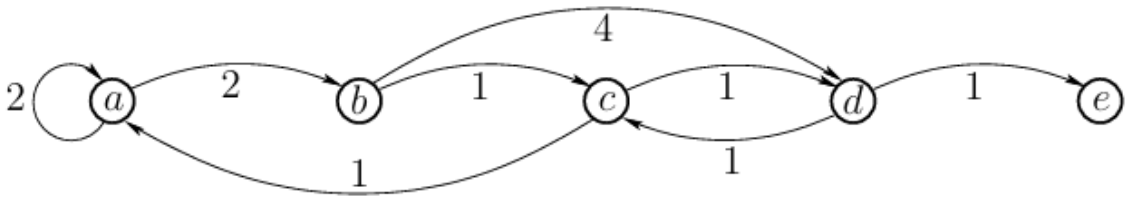


Figure 2.8: A five-state example. Each vertex represents a state, and each edge represents an input that can be applied to the state transition equation to change the state. The weights on the edges represent $l(x_k, u_k)$ (x_k is the originating vertex of the edge).

(a)

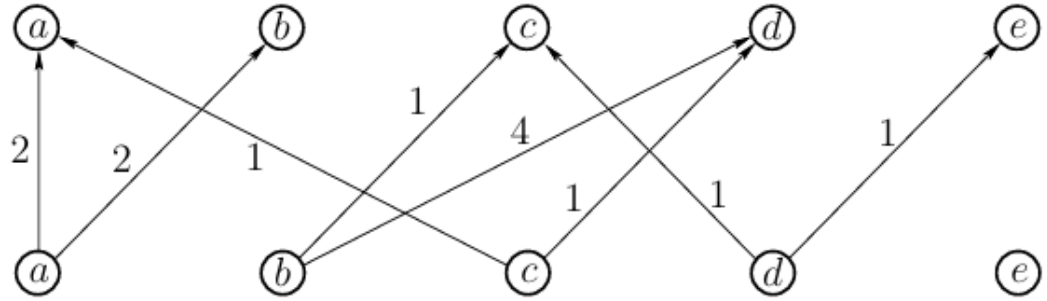


Figure 2.10: The possibilities for advancing forward one stage. This is obtained by making two copies of the states from Figure 2.8, one copy for the current state and one for the potential next state.

(b)

Naive solution

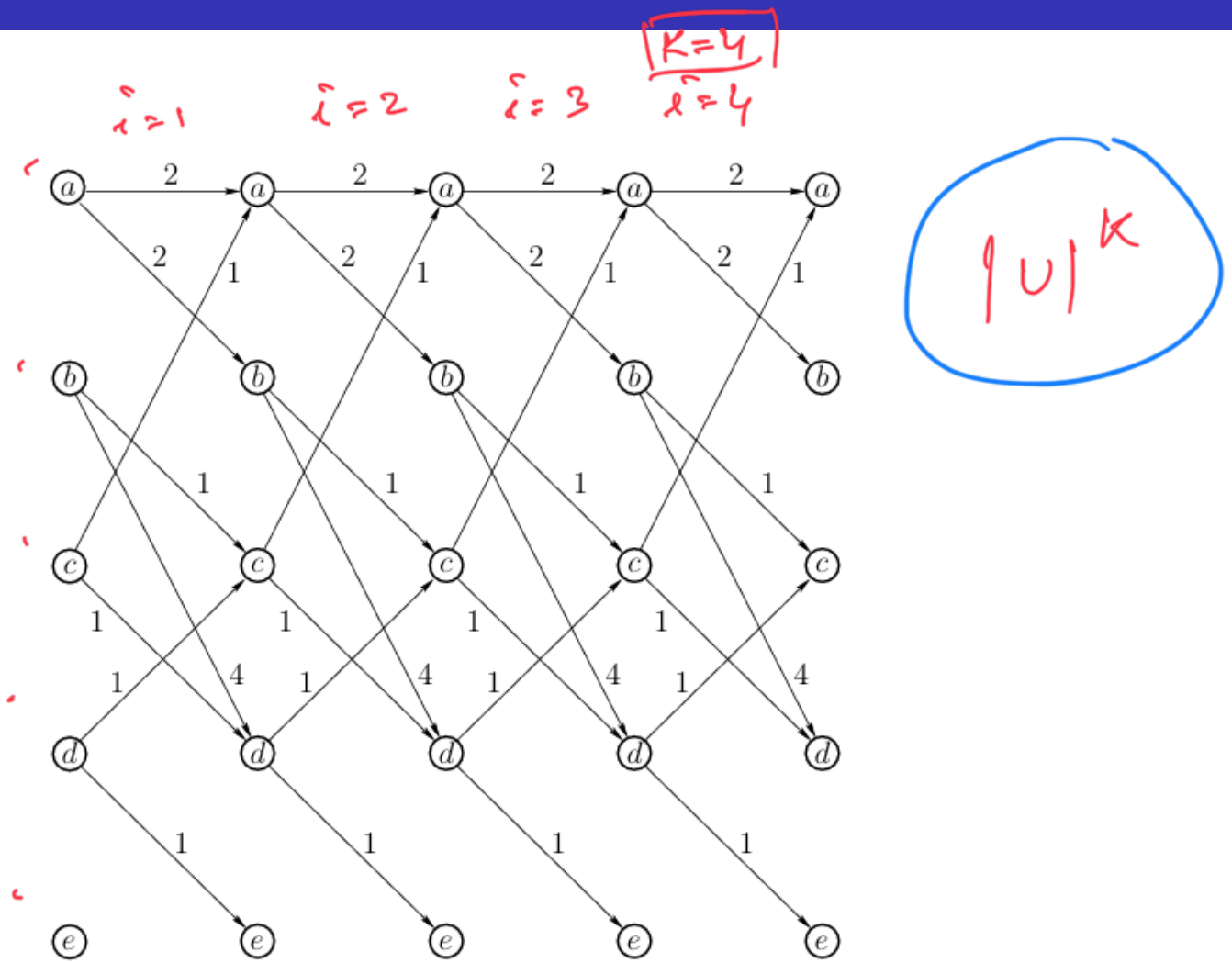


Figure 2.11: By turning Figure 2.10 sideways and copying it K times, a graph can be drawn that easily shows all of the ways to arrive at a final state from an initial state by flowing from left to right. The computations automatically select the optimal route.

Backward value iteration – example

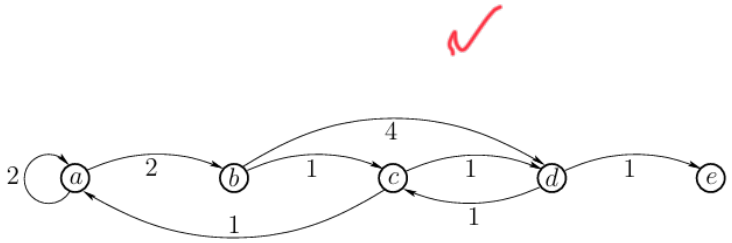


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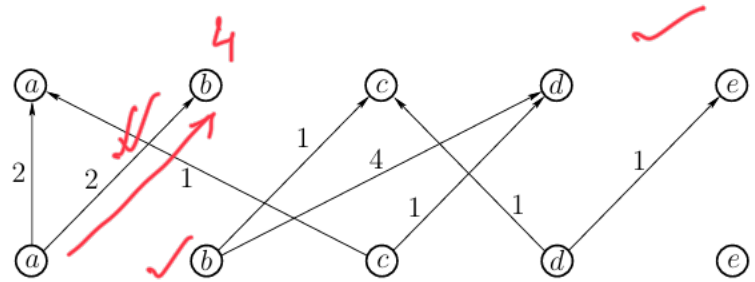


Figure 2.10: The possibilities for advancing forward one stage. This is obtained by making two copies of the states from Figure 2.8, one copy for the current state and one for the potential next state.

(a)

(b)

From a, with 4 actions you can reach X_G with cost 6.

$x_G = \{d\}$

$x_I = a$.

	a	b	c	d	e
G_5^*	∞	∞	∞	0	∞
G_4^*	∞	4	1	∞	∞
G_3^*	6	2	∞	2	∞
G_2^*	4	6	3	∞	∞
G_1^*	6	4	5	4	∞

$G_4^*(b) = 4$??

$G_3^*(a) = 6$

Figure 2.9: The optimal cost-to-go functions computed by backward value iteration.

(c)

Backward value iteration – example

For K-length sequence (i.e., K iterations)

$$O\left(|U| \cdot |X| \cdot K\right)$$

∞ value of $G_i^*(x_i)$ — optimally

- A goal state from x_i is not reachable with $K-i+1$ action sequence
  - $\nexists\, u_i \in U(x_i)$ that takes x_i to a state x_{i+1} from which a $(K-i)$-step plan exists that optimally terminate at $\boxed{X_{G_2}}$.

Iteration proceeds

$$G_F^* \to G_K^* \to G_{K-1}^* \cdots G_i^* \to G_{i-1}^* \cdots G_2^* \to G_1^*$$

Backward value iteration – example

Forward Value Iteration

Forward value iteration

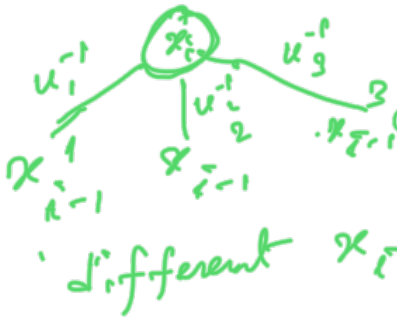
- X_G is not an essential information at the beginning of the algorithm.
- x_I must be fixed at the beginning of the algorithm.
- Backward value iteration: Finds optimal plans from all states simultaneously.
- Forward value iteration: Finds optimal plans to all states in X from x_I.

Forward value iteration

For an intermediate stage $i \in \{2, \ldots, K\}$

$$c_i^*(x_i) = \min_{u_1 \ldots u_{i-1}} \left\{ l_I(x_1) + \sum_{j=1}^{i-1} l(x_j, u_j) \right\}$$

$l_I(x_1)$ → no action



different x_{i-1}'s

$c_1^*(x_1) = l_I(x_1) \rightarrow \begin{cases} l_I(x_I) = 0 \\ = \infty \text{ if } x_1 \neq x_I \end{cases}$

$$c_K^*(x_f) = \min_{u_1 \ldots u_K} \left\{ l_I(x_1) + \sum_{j=1}^{K} l(x_j, u_j) \right\}$$

fix.

$$* \quad c_i^*(x_i) = \min_{u^{-1} \in U^{-1}(x_i)} \left\{ c_{i-1}^*(x_{i-1}) + l(x_{i-1}, u_{i-1}) \right\}$$

$x_i = f(x_{i-1}, u_{i-1})$

$g(x_i, 0) \rightarrow x_{i-1}$

$f^{-1}(x_i, u_i^{-1}) \rightarrow x_{i-1}$

Forward value iteration – example

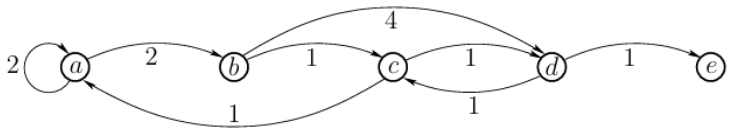


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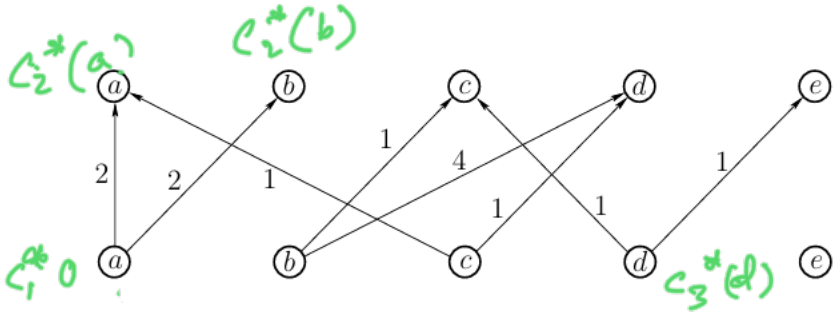


Figure 2.10: The possibilities for advancing forward one stage. This is obtained by making two copies of the states from Figure 2.8, one copy for the current state and one for the potential next state.

(a)

	a	b	c	d	e
C_1^*	0	∞	∞	∞	∞
C_2^*	2	2	∞	∞	∞
C_3^*	4	4	3	6	∞
C_4^*	4	6	5	4	7
C_5^*	6	6	5	6	5

(b)

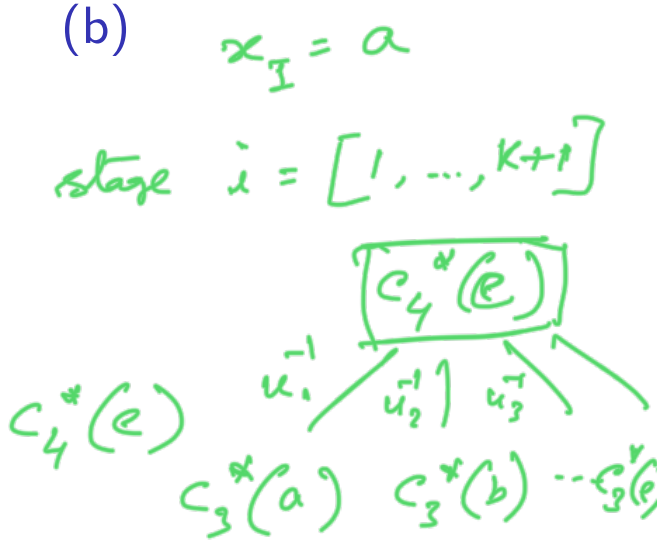
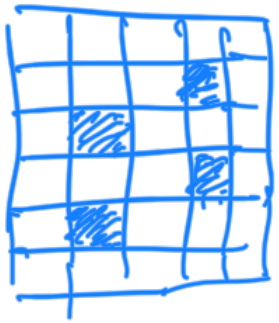


Figure 2.12: The optimal cost-to-come functions computed by forward value iteration.

* C_i^* or G_i^* calculation : The values for all nodes in the (c) state space is computed.

Forward value iteration – example

- Workspace is given [e.g., grid-space]
- Robot motion dynamics is given [e.g., E-W-N-S, 60 motion primities]



How to create a graph from the above inputs?

Value iteration

Thank you